

- 1 The enzyme catalase catalyses the reaction shown in Fig. 1.1.

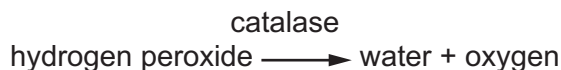


Fig. 1.1

A student had read that copper(II) sulfate can act as an inhibitor, reducing the activity of catalase.

The student wanted to investigate the effect of copper(II) sulfate as an inhibitor on the activity of catalase to test the hypothesis:

The higher the concentration of the inhibitor copper(II) sulfate, the lower the volume of oxygen produced in the presence of catalase.

- (a) State the independent variable **and** dependent variable in this investigation.

independent variable

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dependent variable

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[2]

- (b) (i) The student was provided with a 0.04 mol dm^{-3} solution of copper(II) sulfate.

Describe how the student could use serial dilution to make a suitable range of concentrations to test their hypothesis.

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..... [2]

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- 5 cm³ of 10% hydrogen peroxide
- mixture of catalase and copper(II) sulfate
- graduated tube to collect oxygen
- water

Describe a method the student could use to collect the data needed to test the effect of different concentrations of copper(II) sulfate on the production of oxygen in the presence of catalase.

You should **not** repeat the details from **(b)(i)** describing how to dilute the 0.04 mol dm^{-3} copper(II) sulfate solution or how to set up the apparatus shown in Fig. 1.2.

[illegible]

[6]

(iii) The hypothesis the student tested was:

The higher the concentration of the inhibitor copper(II) sulfate, the lower the volume of oxygen produced in the presence of catalase.

Complete Fig. 1.3 by:

- adding axes labels and units
- sketching a graph of the results you would expect if the hypothesis is correct.



Fig. 1.3

(c) The student carried out a further investigation to find out if copper(II) sulfate was a competitive or non-competitive inhibitor.

- The student measured the oxygen produced in the presence and absence of copper(II) sulfate at different concentrations of hydrogen peroxide.
- The student processed the data to find the **rate** of oxygen production.

The student then plotted these rates on the graph shown in Fig. 1.4.

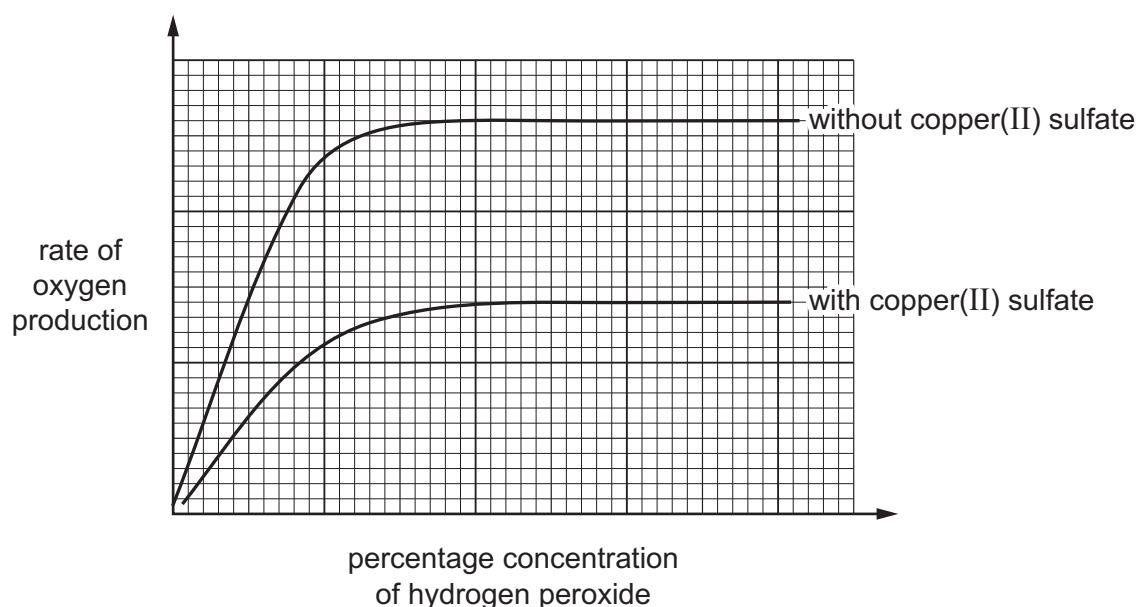


Fig. 1.4

- (i) Draw on Fig. 1.4 to show how you could derive the K_m value of catalase **without** copper(II) sulfate. [3]
- (ii) The student concluded that copper(II) sulfate was a non-competitive inhibitor of catalase.

State the evidence in Fig. 1.4 that supports this conclusion.

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